Assignment 4: Matlab MA 122 07/02/25
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1. Plot x versus y for $y = \sin(x)$. Let x vary from 0 to 2π in increments of 0.1π.
   (a) Add a title and labels to your plot.

2. Plot x versus $y1$ and $y2$ for $y1 = \sin(x)$ and $y2 = \cos(x)$. Let x vary from 0 to 2π in increments of 0.1π. Add a title and labels to your plot
   (a) Add a legend to the graph.
   (b) Adjust the axes so that the x-axis goes from -1 to $2\pi + 1$ and the y-axis from -1.5 to $+1.5$.

3. Subdivide a figure window into two rows and one column.
   (a) plot $y = \tan(x)$ for $-1.5 \leq x \leq 1.5$. Use an increment of 0.1.
   (b) Add a title and axis labels to your graph.
   (c) plot $y = \sinh(x)$ for the same range.
   (d) Add a title and labels to your graph.

4. Create a plot of the functions that follow, using **fplot**.
   (a) $f(t) = 5t^2$.
   (b) $f(t) = 5(\sin(t))^2 + t(\cos(t))^2$.
   (c) $f(t) = \sin(t) + \log(t)$.

5. Create a vector x of values from 0 to 20π, with a spacing of $\frac{\pi}{100}$. Define vectors y and z as

   $$y = x\sin(x)$$

   and

   $$z = x\cos(x)$$

   (a) Create an x–y plot of x and y.
   (b) Create a polar plot of x and y.
   (c) Create a three-dimensional line plot of x, y, and z. Don't forget a title and labels.
   (d) Generate a contour plot of Z.
   (e) Generate a combination surface and contour plot of Z.

6. Create x and y vectors from -5 to $+5$ with a spacing of 0.5. Use the meshgrid function to map x and y onto two new two-dimensional matrices called X and Y. Use your new matrices to calculate vector Z, with magnitude

   $$Z = \sin(\sqrt{X^2 + Y^2})$$

   (a) Use the mesh plotting function to create a three-dimensional plot of Z.
   (b) Use the surf plotting function to create a three-dimensional plot of Z. Compare the results you obtain with a single input Z with those obtained with inputs for all three dimensions (X, Y, Z).

7. Let the vector $G = [68, 83, 61, 70, 75, 82, 57, 5, 76, 85, 62, 71, 96, 78, 76, 68, 72, 75, 83, 93]$ represent the distribution of final grades in an engineering course.
   (a) Use MATLAB to sort the data and create a bar graph of the scores.
   (b) Create a histogram of the scores, using hist.

8. Draw the mesh plot of following function:
   (a) $z = \frac{xe^x}{2y}$ for $1 \leq x \leq 5$ and $1 \leq y \leq 5$.
   (b) $z = \log(x^2 + y^2)$ for $1 \leq x \leq 5$ and $1 \leq y \leq 5$.
   (c) $z = \frac{y^P 2}{x^2+y^2}$ for $1 \leq x \leq 5$ and $1 \leq y \leq 5$